Project Documentation & User Manual: Installation, Operations & Administration

Project: NeuroClass — Multimodal Attendance & Engagement Agent

Module: `7.Project Documentation`

Phase: System Documentation, User Manual & Troubleshooting Guide

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1. System Overview & Architecture Highlights

NeuroClass is an offline edge AI system designed to eliminate manual attendance, quantify classroom engagement through privacy-safe vision and acoustic metrics, and provide live pedagogical nudges.

NEUROCLASS USER INTERFACE LANDSCAPE	
1. REAL-TIME HUD DASHBOARD (Port 8000)	2. STREAMLIT ANALYTICS COCKPIT (Port 8501)
- Live video feed with facial privacy blur	- Longitudinal session attendance heatmaps
- Real-time attendance count (Present / Late)	- Attentive vs. Distracted student breakdowns
- 3-Way Hybrid Student Enrollment interface	- Detailed focus zone curves & percentile stats
- Active AI pedagogical toasts (45s cooldown)	- Official tamper-evident attendance CSV export

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2. Step-by-Step Installation & Environment Setup

Follow these procedural instructions to configure the Python virtual environment, install validated dependencies, and prepare the local SQLite database.

Step 1: Clone Repository & Create Virtual Environment

Open PowerShell or your preferred terminal in the project directory:

```
# Verify Python version (3.10+ required; tested on 3.14.8)
python --version

# Create an isolated virtual environment
python -m venv .venv

# Activate the virtual environment
# Windows PowerShell:
.\.venv\Scripts\Activate.ps1
# Linux / macOS:
source .venv/bin/activate
```

Step 2: Install Required Dependencies

Install the verified 14 core dependencies and runtime enhancements:

```
cd "5. Project Development Phase"
pip install -r requirements.txt
```

Step 3: Validate the Development Environment

Run the automated pre-flight verification script to ensure all 14 libraries and system modules load cleanly:

```
python validate_environment.py
```

Expected output:

```
================================================================================
       NEUROCLASS: EPIC 1 - ACTIVITY 1.2 ENVIRONMENT VERIFICATION REPORT
================================================================================
  [VERIFIED] fastapi               : 0.143.0
  [VERIFIED] opencv-python         : 5.0.0
  [VERIFIED] mediapipe             : 1.1.0
```

```
  ...
[SUCCESS] ALL 14 CORE LIBRARIES + ENHANCEMENTS ARE FULLY IMPLEMENTED AND VERIFIED!
================================================================================
```

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3. Operational Manual: Running the Core Application

Once installed, NeuroClass provides flexible execution modes allowing administrators to launch the complete system either from the root forwarder or directly within the project development phase.

Option A: Running from Repository Root (Recommended)

You can launch any component directly from the root workspace using the universal forwarder:

```
# 1. Execute end-to-end 5-module pipeline verification
python run_from_root.py --verify

# 2. Run the 91-test verification suite
python run_from_root.py --test

# 3. Launch the core FastAPI server &amp;amp; Web HUD
python run_from_root.py --agent

# 4. In a separate terminal, launch the Streamlit analytics cockpit
python run_from_root.py --streamlit
```

Option B: Running inside `5. Project Development Phase`

```
cd "5. Project Development Phase"

# Launch FastAPI ASGI Server (Default port: 8000)
python run_agent.py

# Optional CLI flags:
python run_agent.py --source webcam --webcam-index 0   # Use physical USB webcam
python run_agent.py --source simulated                 # Use automated simulated classroom feed
python run_agent.py --source file --video-path path.mp4 # Ingest pre-recorded lecture video
```

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4. User Guide: The 3-Way Hybrid Biometric Enrollment

NeuroClass ensures that missing student photos never delay a live classroom:

HOW TO ENROLL STUDENTS INTO NEUROCLASS
METHOD 1: File Directory Drop
• Place student portrait photos directly into: `5. Project Development Phase/student_db/`
• File naming format: `STU-101.jpg` or `STU-101_Emma_Watson.jpg`
• Restart server or refresh registry: Embeddings are auto-calculated and cached on startup.
METHOD 2: Web Portal Upload
• Open the Web HUD at `http://127.0.0.1:8000` and navigate to the Student Management drawer.
• Enter Student ID, Student Name, and upload a JPG/PNG portrait crop. Click "Register Face".
METHOD 3: Live 1-Click Track Linking ([IDENTIFY / LINK] Identify / Link)
• When an unverified participant is detected, they appear on the HUD with ID: `TRK-001`.
• Click the "[IDENTIFY / LINK] Identify / Link" button next to their bounding box.
• Enter their official Student ID (e.g., `STU-109`). The system crops their face instantly,
saves it to `student_db/STU-109.jpg`, and updates all attendance records automatically!

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5. Live Classroom Operations & Pedagogical HUD

1. Starting a Classroom Session:

• In the Web HUD, enter the **Course Name** (e.g., *"Advanced Multimodal AI"*) and **Instructor Name**.

• Click **Start Session**. The timer begins, and the WebSocket telemetry stream initiates.

2. Monitoring Active Presence:

• The HUD displays the live participant grid.

• Verified students display a green badge (`STU-101 • Emma Watson • 94% Focus`).

• Transient walk-bys (***<15***s) are automatically filtered out.

3. Observing Pedagogical Alerts:

• If the instructor lectures continuously without student participation for ***>12*** minutes, a toast notification appears:

> *"Pacing Alert: 12-minute monologue detected. Consider asking a check-in question."*

• Alerts adhere to a strict 45-second cooldown to eliminate notification fatigue.

4. Concluding Session & Exporting Data:

• Click **End Session**.

• Navigate to `http://127.0.0.1:8501` (Streamlit) or click **Export CSV** on the Web HUD.

• The generated CSV file includes:

• `Student ID`, `Student Name`, `First Seen Timestamp`, `Last Seen Timestamp`

• `Active Presence Minutes`, `Attendance Status` (*Present*, *Late*, *Absent*)

• `Average Engagement Index (%)`

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6. Troubleshooting & Diagnostics

Symptom	Probable Cause	Corrective Action	
`FastAPI Server : OFFLINE`	Server not yet started on port 8000	Launch `python run_from_root.py --agent` in a terminal.	
`Camera index 0 unavailable`	Another application is holding the camera handle	Close Zoom/Teams or test simulated mode: `python run_agent.py --source simulated`.	
`Missing ScriptRunContext` warning	Harmless Streamlit bare-mode log during automated tests	Normal unit test behavior; can be safely ignored.	
`ModuleNotFoundError: backend`	Python invoked outside proper path context	Run through `run_from_root.py` or run `cd "5. Project Development Phase"` first.	
`Port 8000 already in use`	Previous uvicorn server still running in background	Kill existing process via PowerShell: `Get-Process python \`	Stop-Process -Force`.
`Microphone permission denied`	Browser audio capture blocked	Allow microphone permissions in browser settings for `http://127.0.0.1:8000`.	